

1. Network Topology & Architecture

The network consists of a redundant triangle topology containing three switches (**S1**, **S2**, **S3**) interconnected via Gigabit Ethernet trunks.

Redundancy & Loops: Having three switches connected in a physical loop provides hardware redundancy. However, without Spanning Tree Protocol (STP), this layout would trigger catastrophic broadcast storms and MAC table instability.

STP Behavior: To prevent loops, STP elects one **Root Bridge**. The other switches find their lowest-cost path to the Root. Any redundant paths that could form a loop are placed into a **Blocking (BLK)** state, disabling data forwarding over that specific link.

2. Complete Device Configurations

Because this lab focuses on observing the default behavior of Spanning Tree Protocol, the primary configuration requirements involve enabling standard trunking on the Gigabit interfaces and ensuring basic VLAN 1 connectivity.

Switch 1 (S1) Configuration

```
S1> enable
S1# configure terminal

! Set up trunk interfaces to S2 and S3
S1(config)# interface range g0/1 - 2
S1(config-if-range)# switchport mode trunk
S1(config-if-range)# no shutdown
S1(config-if-range)# exit

! Assign IP address for management on default VLAN 1
S1(config)# interface vlan 1
S1(config-if)# ip address 192.168.1.11 255.255.255.0
S1(config-if)# no shutdown
S1(config-if)# exit
```

Switch 2 (S2) Configuration

```
S2> enable
S2# configure terminal

! Set up trunk interfaces to S1 and S3
```

```
S2(config)# interface range g0/1 - 2
S2(config-if-range)# switchport mode trunk
S2(config-if-range)# no shutdown
S2(config-if-range)# exit

! Host access port configurations for PC1
S2(config)# interface FastEthernet0/1
S2(config-if)# switchport mode access
S2(config-if)# switchport access vlan 1
S2(config-if)# no shutdown
S2(config-if)# exit

! Management IP
S2(config)# interface vlan 1
S2(config-if)# ip address 192.168.1.12 255.255.255.0
S2(config-if)# no shutdown
S2(config-if)# exit
```

Switch 3 (S3) Configuration

```
S3> enable
S3# configure terminal

! Set up trunk interfaces to S1 and S2
S3(config)# interface range g0/1 - 2
S3(config-if-range)# switchport mode trunk
S3(config-if-range)# no shutdown
S3(config-if-range)# exit

! Host access port configurations for PC2
S3(config)# interface FastEthernet0/1
S3(config-if)# switchport mode access
S3(config-if)# switchport access vlan 1
S3(config-if)# no shutdown
S3(config-if)# exit

! Management IP
S3(config)# interface vlan 1
S3(config-if)# ip address 192.168.1.13 255.255.255.0
S3(config-if)# no shutdown
S3(config-if)# exit
```

3. Reference Addressing Table

(Note: Default flat management network configuration assumed based on standard Cisco Academy baseline).

Device	Interface	IP Address	Subnet Mask	Connected To
S1	VLAN 1	192.168.1.11	255.255.255.0	Core / Trunking Node
S2	VLAN 1	192.168.1.12	255.255.255.0	Distribution Node
S3	VLAN 1	192.168.1.13	255.255.255.0	Distribution Node
PC1	NIC	192.168.1.21	255.255.255.0	S2 (\$F0/1\$)
PC2	NIC	192.168.1.22	255.255.255.0	S3 (\$F0/1\$)

4. Spanning Tree State Analysis & Answers

Part 1: Converged Instance State Table

Based on standard Spanning Tree structural elections where S1 holds the lowest MAC address/highest priority (Root Bridge):

Switch	Port	Status	Root Bridge?
S1	\$G0/1\$	FWD (Forwarding)	Yes
S1	\$G0/2\$	FWD (Forwarding)	Yes
S2	\$G0/1\$	FWD (Forwarding)	No
S2	\$G0/2\$	FWD (Forwarding)	No
S3	\$G0/1\$	FWD (Forwarding)	No
S3	\$G0/2\$	BLK (Blocking)	No

Guided Questions Answered

What does the orange link light mean? It indicates that the port is in a non-forwarding state—specifically, the **Blocking (BLK)** state enforced by STP.

What path will frames take from PC1 to PC2? Frames travel from **PC1** $\rightarrow$ **S2** $\rightarrow$ **S1** $\rightarrow$ **S3** $\rightarrow$ **PC2**.

Why do the frames not travel through S3 directly? The direct path between S2 and S3 contains the port placed in a blocking state (G0/2 on S3) to kill the logical routing loop.

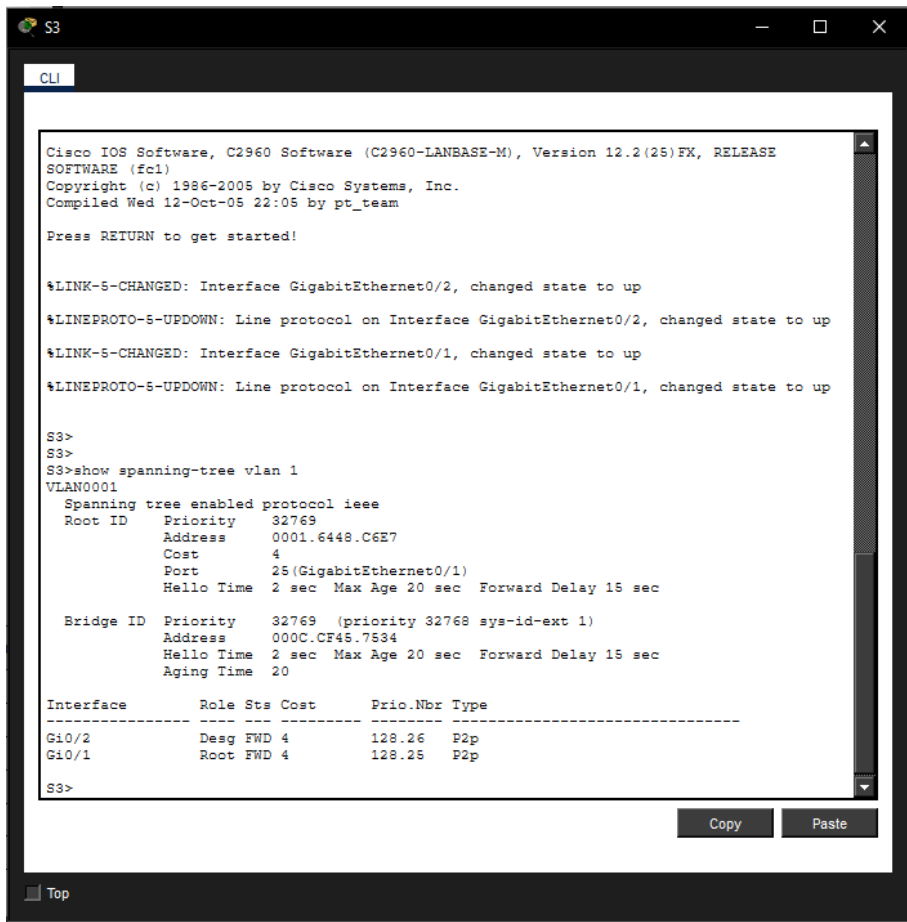
Why has spanning tree placed a port in a blocking state? To prevent data frames from continuously looping around the redundant path triangle. Layer 2 Ethernet headers lack a Time-to-Live (TTL) counter; without STP blocking a port, broadcast frames would loop forever and crash the network.

5. Verification & Convergence Playbook

Step A: Verify Initial Status

Check the convergence states on S3 using:

`S3# show spanning-tree vlan 1`

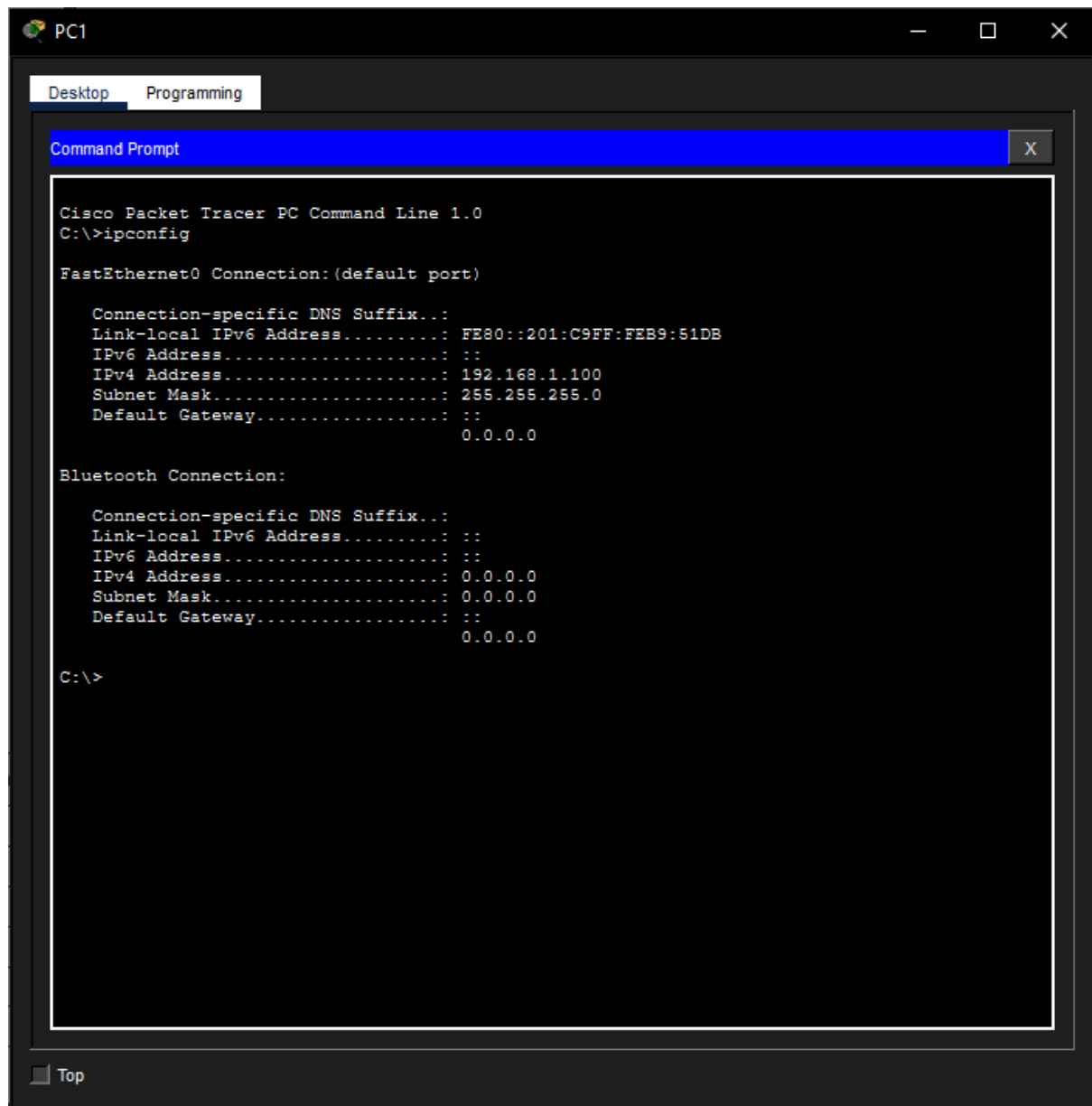


Expected Output Snippet:				Interface	Role	Stp	Cost	Status	Type
Gi0/1	Root	FWD	4	P2p	Gi0/2	Altn	BLK	4	P2p

Step B: Host Baseline Validation

On **PC1**, verify the local interface configuration:

```
C:\> ipconfig
```



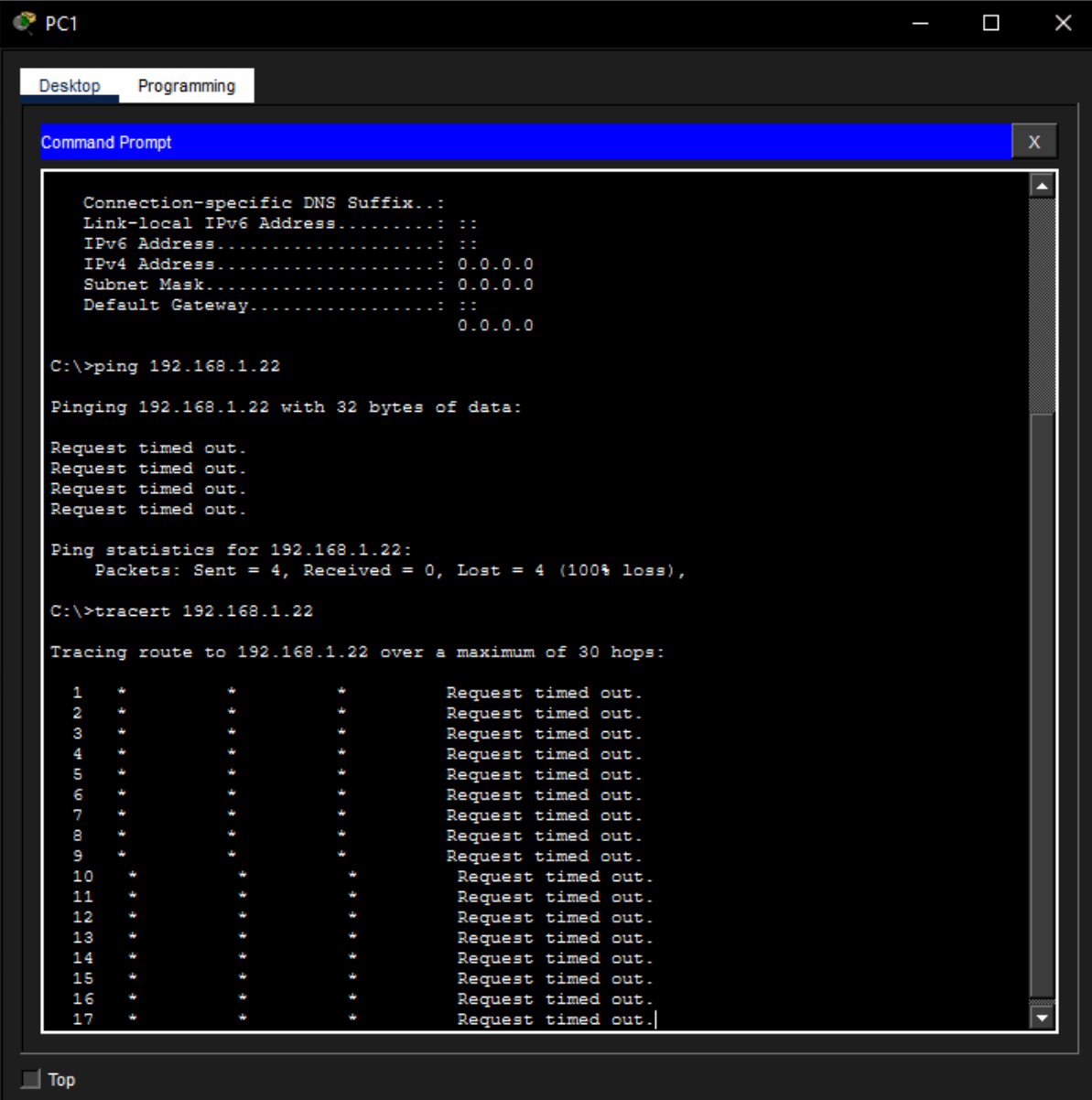
Verify path transparency through the loop prevention mechanism by running a ping and traceroute:

```
C:\> ping 192.168.1.22
```

Result: Reply from 192.168.1.22: bytes=32 time<1ms TTL=128

C:\> tracert 192.168.1.22

Result: Resolves across the standard bridged path via S1 without looping.



The screenshot shows a Windows desktop environment with a taskbar at the top. The active window is titled 'PC1' and contains a 'Command Prompt' application. The Command Prompt window has a blue title bar and a black background with white text. The text in the Command Prompt shows the output of several network commands: 'ipconfig' (showing IPv4 address 0.0.0.0), 'ping 192.168.1.22' (showing 100% packet loss), and 'tracert 192.168.1.22' (showing 17 hops, all with 'Request timed out.'). The Command Prompt window is positioned over a 'Programming' tab in the background. At the bottom left of the PC1 desktop, there is a 'Top' button.

```
PC1
Desktop Programming
Command Prompt
Connection-specific DNS Suffix...:
Link-local IPv6 Address...: ::
IPv6 Address...: ::
IPv4 Address...: 0.0.0.0
Subnet Mask...: 0.0.0.0
Default Gateway...: ::
0.0.0.0

C:\>ping 192.168.1.22

Pinging 192.168.1.22 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 192.168.1.22:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>tracert 192.168.1.22

Tracing route to 192.168.1.22 over a maximum of 30 hops:
  0  *          *          *          Request timed out.
  1  *          *          *          Request timed out.
  2  *          *          *          Request timed out.
  3  *          *          *          Request timed out.
  4  *          *          *          Request timed out.
  5  *          *          *          Request timed out.
  6  *          *          *          Request timed out.
  7  *          *          *          Request timed out.
  8  *          *          *          Request timed out.
  9  *          *          *          Request timed out.
 10 *          *          *          Request timed out.
 11 *          *          *          Request timed out.
 12 *          *          *          Request timed out.
 13 *          *          *          Request timed out.
 14 *          *          *          Request timed out.
 15 *          *          *          Request timed out.
 16 *          *          *          Request timed out.
 17 *          *          *          Request timed out.

Top
```

Step C: Observe Convergence Loop Failover

Repeatedly polling S3 commands shows the classic structural sequence:

-
- The screenshot shows the Packet Tracer interface with the following components:
- CLI Window (S3):**

```

S3#show spanning-tree vlan 1
VLAN0001
Spanning tree enabled protocol ieee
Root ID    Priority    32769
Address    0001.6440.C627
Cost       4
Port       28 (GigabitEthernet0/1)
Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec

Bridge ID  Priority    32769 (priority 32769 sys-id-ext 1)
Address    000C.CF43.7834
Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec
Aging Time 20

Interface Role Sts Cost Prio.Nbr Type
-----
Gi0/2 Desg FWD 4 128.26 P2p
Gi0/1 Root FWD 4 128.26 P2p

S3#show spanning-tree vlan 1
VLAN0001
Spanning tree enabled protocol ieee
Root ID    Priority    32769
Address    0001.6440.C627
Cost       4
Port       28 (GigabitEthernet0/1)
Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec

Bridge ID  Priority    32769 (priority 32769 sys-id-ext 1)
Address    000C.CF43.7834
Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec
Aging Time 20

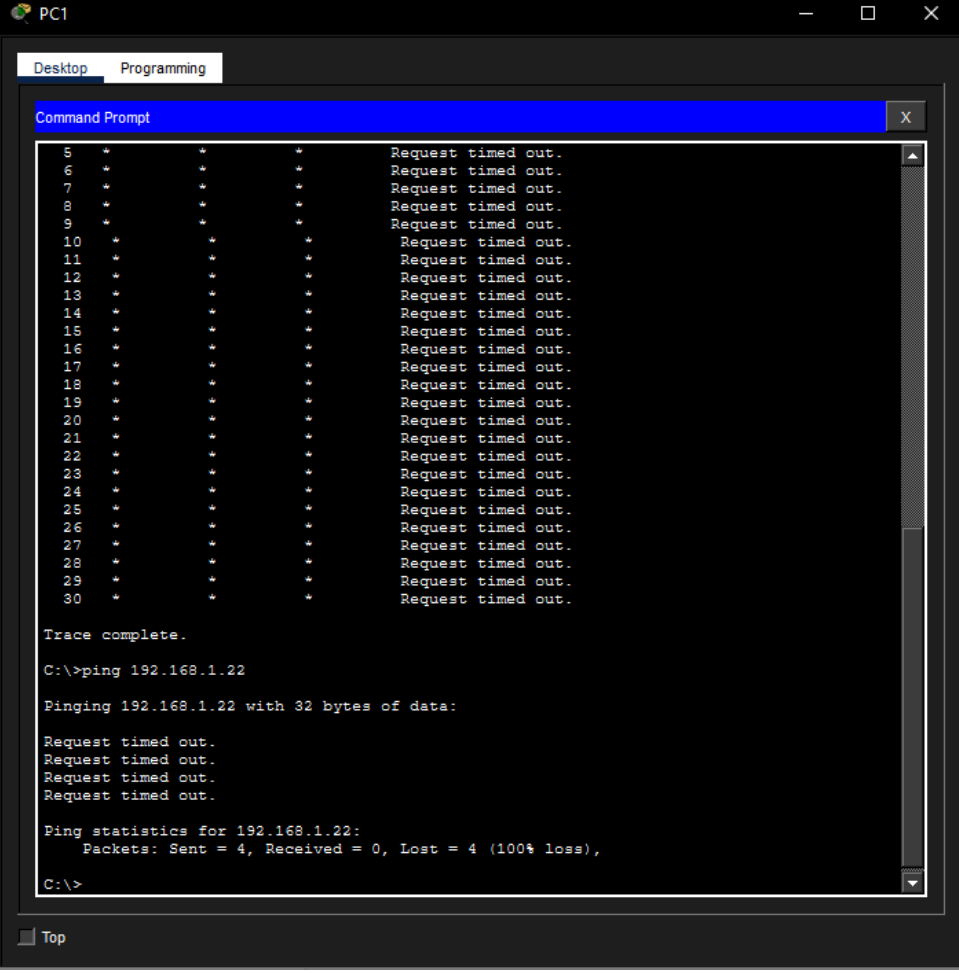
Interface Role Sts Cost Prio.Nbr Type
-----
Gi0/2 Desg FWD 4 128.26 P2p
Gi0/1 Root FWD 4 128.26 P2p

S3#

```
 - Network Diagram:**
 - Switch S3 is connected to Switch S2 via G0/1.
 - Switch S2 is connected to Switch S1 via G0/1.
 - Switch S3 is connected to Switch S1 via G0/2.
 - PC1 (192.168.1.100) is connected to S1.
 - PC2 (192.168.1.101) is connected to S2.
 - Bottom Panel:**
 - Time: 00:15:44
 - Scenario 0
 - Buttons: New, Delete, Toggle PCUI Last Window
 - Simulation status: Realtime

FWD (Forwarding): Fully operational port passing data frames.

C:\> ping 192.168.1.22



The screenshot shows a Windows PC1 window with a 'Programming' tab active. Inside is a 'Command Prompt' window. The output of the 'ping 192.168.1.22' command is as follows:

```
5 * * * Request timed out.
6 * * * Request timed out.
7 * * * Request timed out.
8 * * * Request timed out.
9 * * * Request timed out.
10 * * * Request timed out.
11 * * * Request timed out.
12 * * * Request timed out.
13 * * * Request timed out.
14 * * * Request timed out.
15 * * * Request timed out.
16 * * * Request timed out.
17 * * * Request timed out.
18 * * * Request timed out.
19 * * * Request timed out.
20 * * * Request timed out.
21 * * * Request timed out.
22 * * * Request timed out.
23 * * * Request timed out.
24 * * * Request timed out.
25 * * * Request timed out.
26 * * * Request timed out.
27 * * * Request timed out.
28 * * * Request timed out.
29 * * * Request timed out.
30 * * * Request timed out.

Trace complete.

C:\>ping 192.168.1.22

Pinging 192.168.1.22 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 192.168.1.22:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>
```

Post-Convergence Result: Successful link recovery! There are no longer any orange link lights because the network has lost its redundancy, eliminating any potential physical loops.

6. Personal Reflections

- **The Silent Hero of Layer 2:** This activity shows that STP is an essential protocol for maintaining network stability. It operates completely in the background, automatically managing link changes without requiring manual admin intervention.

จากการทำแล็บนี้ทำให้เข้าใจว่า Spanning Tree Protocol (STP) เป็นโปรโตคอลที่สำคัญอย่างยิ่งในการรักษาเสถียรภาพของระบบเครือข่าย โดยระบบจะทำงานอยู่เบื้องหลังโดยอัตโนมัติเพื่อจัดการการเปลี่ยนแปลงของเส้นทางเชื่อมต่อเมื่อเกิดเหตุการณ์สายสัญญาณขาด ทำให้ระบบเครือข่ายทำงานต่อได้โดยไม่ต้องอาศัยการแก้ไขโครงสร้างจากผู้ดูแลระบบด้วยตนเอง

- **The Cost of Convergence:** Watching the interface transition through Listening and Learning phases underscores the inherent slowness of legacy STP (802.1D), which can take up to 30–50 seconds to fail over. In modern production environments, replacing this with Rapid Spanning Tree Protocol (RSTP / 802.1w) is critical to reduce convergence delays down to milliseconds.

การได้สังเกตกระบวนการเปลี่ยนผ่านสถานะของพอร์ตสวิตช์ผ่านขั้นตอน Listening และ Learning ทำให้เห็นถึงข้อจำกัดด้านความล่าช้าของโปรโตคอล STP รุ่นเก่า (802.1D) ซึ่งต้องใช้เวลาฟื้นตัวนานถึง 30–50 วินาที ส่งผลให้ทราบว่าในระบบเครือข่ายระดับองค์กรจริงปัจจุบันควรเปลี่ยนมาใช้งาน Rapid Spanning Tree Protocol (RSTP / 802.1w) เพื่อลดระยะเวลาการฟื้นตัวของระบบให้เหลือเพียงระดับมิลลิวินาที เพื่อประสิทธิภาพและความต่อเนื่องสูงสุดของเครือข่าย

643020651-7 นางสาวสุชานาฏ แก้วมุงคุณ